
26 ASL Alphabet Handshapes Classification

Linfan (Frank) Bao

1011102511, frank.bao@mail.utoronto.ca

https://github.com/linfan-bao/aps360_asl.git

Introduction

Sign language is a primary means of communication for the Deaf community, but access to fingerspelling interpretation services is still limited. Motivated by a Government of Canada report on gaps in accessibility services for the Deaf community [2], we believe fingerspelling-recognition technology can improve communication equity. The goal is to construct a residual CNN and a classifier for the ASL 26-alphabet gestures with real-world backgrounds.

Since the mapping from pixels to a letter is hard to capture with fixed rules, a learned machine learning approach is appropriate. We choose deep learning's CNN structure because of its ability to learn hierarchical visual features such as edges, textures and hand-shape patterns. This capability suits the task of recognizing ASL fingerspelling and matters because some letters differ only in subtle finger configurations. As Figure 1 illustrates, our model takes an RGB hand image (resized to 224×224) as input and outputs a probability distribution over the 26 letters. The most likely class is the output prediction.

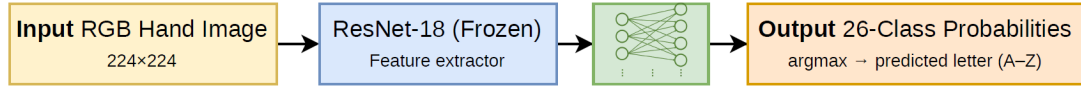


Figure 1: overview of primary model

Data Processing

Our dataset is collected from 2 sources. For training and validation, Kaggle ASL Alphabet dataset [1] is used. It contains 200×200 RGB images of 26 static ASL alphabet handshapes (A–Z; 3000 per class; 78000 in total). For testing, we collect our own images. All images undergo the preprocessing of resizing to 224×224 and normalizing through the ImageNet channel means [0.485, 0.456, 0.406] and standard deviations [0.229, 0.224, 0.225] matching the input distribution of pretrained Res-net. Data augmentation is only applied to the training set for regularization purposes.

The preprocessing pipeline is illustrated by Figure 2. From each class, we randomly subsample 800 images (to keep training tractable under our limited compute while preserving class balance) and apply a 85-15 train-validation split. During augmentation, we avoid horizontal flipping, since mirroring a handshape can alter its meaning.

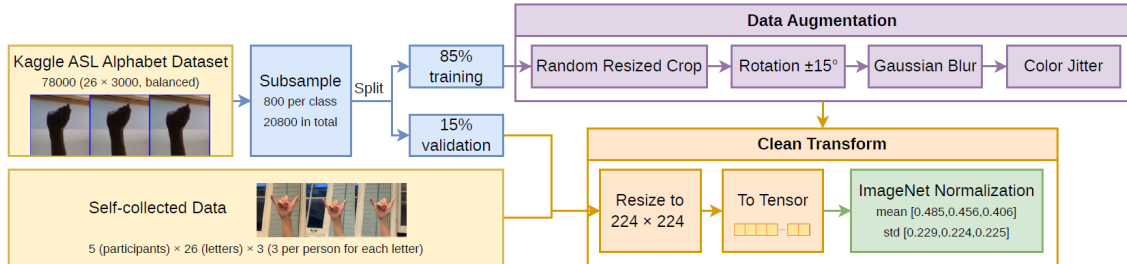


Figure 2: block diagram illustrating procedures for preprocessing of raw data

Figure 3 shows a training sample with its four augmented versions. Despite the random transformations applied, the handshape remains clearly recognizable. This confirms that our augmentation introduces variation without changing the label.

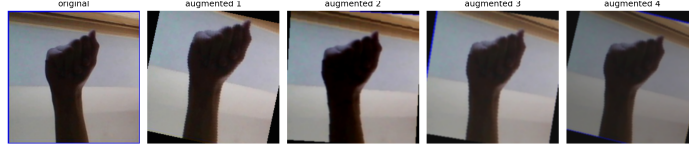


Figure 3: a training sample on the left with four augmented versions

To obtain never-before-seen data, we plan to collect the testing set ourselves. 5 participants each sign all 26 letters, 3 photos per letter with unique lighting, background and camera angle. Only hands and wrists are captured and informed consent will be obtained from every participant.

The main challenge is the domain gap between the clean, single-background Kaggle images and our real-world self-collected photos, which differ in lighting, background, and hand appearance. Bridging this gap is the true test of the model’s practical usefulness.

Baseline Model

For baseline model, we implement a fully-connected (non-convolutional) network which is simple to implement and requires little tuning. As shown in Figure 4, the 224×224 image is flattened and passed through two linear layers (with a ReLU in between) to output probabilities for each letter. All 38.5M parameters are randomly initialized and trainable, with no pretrained modules.

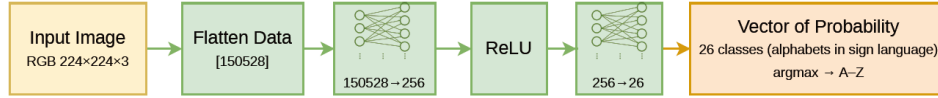


Figure 4: architecture of baseline model

We use the same data and split as described in the Data Processing section. The best validation accuracy was 77.1% (epoch 20) which is far above the 3.8% (1/26) random choice. It is trained with SGD (momentum 0.9), batch size of 32, 20 epochs, learning rate of $1e-4$. Figure 5 shows that the validation accuracy improves steadily and converges to around 77% by epoch 20 (no overfitting). Validation accuracy stays slightly above training since training images are augmented, thus harder to classify.

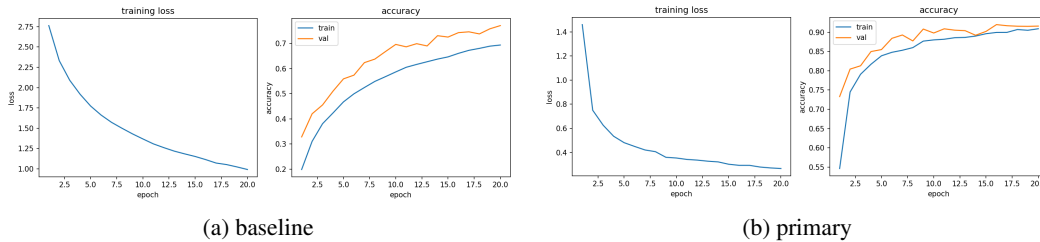


Figure 5: Training and validation curves: (a) baseline vs (b) primary.

Qualitatively, the baseline tends to mix up handshapes “R/U/V/W/X” with each other. ‘U’ has the worst case where it is frequently predicted as “R/V/W” and ‘V’ and ‘X’ are often predicted as ‘R’. These letters share the same basic shape and only differ by how raised fingers are arranged. Moreover, pairs with look-alike gestures such as N-M, P-Q and A-E also appear (less frequent than former ones). These pairs share similar gestures that differ by only one finger’s positioning. Both errors show that the model confuses handshapes that look alike. Distinguishing these patterns will require interpreting spatial relationships between adjacent fingers which baseline’s linear layers are not capable of doing. This gap motivates us to try convolution layers in our primary model.

A challenge we faced was the numerical stability of the baseline. Flattening the image to 150528 dimensional vectors makes the first linear layer enormous. At higher learning rates the training loss diverged. Reducing the learning rate to 1e-4 was necessary to keep training stable, at the cost of slower convergence.

Primary Model

For primary model, we use transfer learning: an ImageNet-pretrained ResNet-18 as a frozen feature extractor followed by a trainable classifier head. As Figure 6 shows, the frozen backbone CNN (with GAP) maps the 224×224 image to a 512-dimensional vector. Then, our classifier head turns it into a probability vector over the 26 letters. Among 11.3M parameters, only 138K (classifier) are trainable, thus reducing training cost and lowering the risk of overfitting.

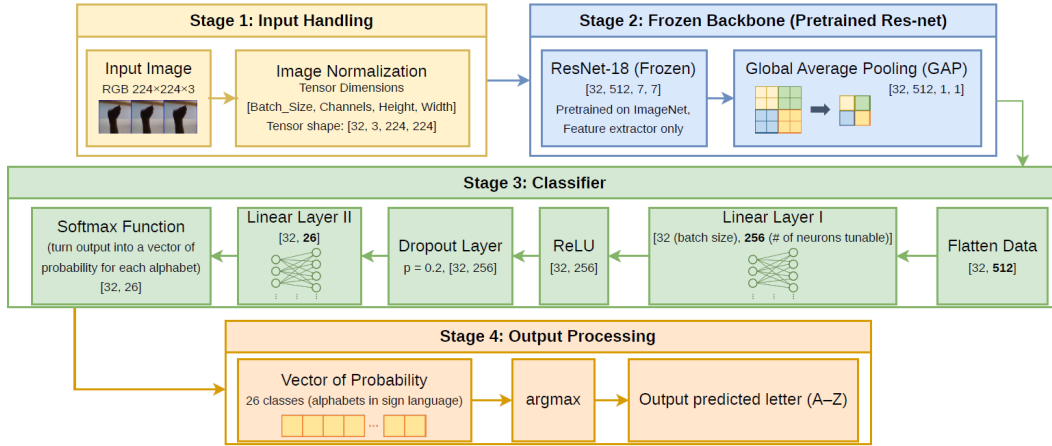


Figure 6: architecture of primary model

We use the same data and split. The best validation accuracy was 92.0% (epoch 16), with a final accuracy of 91.6% which is far higher than both the 3.8% random choice and the 77.1% baseline. It is trained with SGD (momentum 0.9), batch size of 32, 20 epochs (matching the baseline), learning rate of 0.01, dropout of 0.2. Figure 5 shows validation accuracy increasing over 90% by about epoch 10 then plateauing indicating the model converges. As training and validation accuracy remain close, there is no overfitting.

Qualitatively, the primary model makes significantly fewer errors than the baseline. It resolves most of the “R/U/V/W/X” handshape-confusions that the baseline encountered. This shows that the convolutional layers capture the subtle spatial relationships between adjacent fingers that the fully-connected baseline could not. Its main confusions concentrate in handshapes with identical base gestures but different orientations or positions of the raised fingers: R-U, V-K, V-U and I-J. ‘I’ and ‘J’ share the exact same handshape and differ only in orientation which is difficult to distinguish from a single static image, so this error is semantically sensible rather than random.

A challenge we faced during the training stage was an initial underfitting due to a higher dropout of 0.5 and fewer epochs. The validation accuracy was around 80%. Reducing dropout to 0.2 and training for the full 20 epochs let the classifier head converge and achieve the best result.

Table 1 compares the performance of our baseline and primary model. The primary model outperforms the baseline and both exceed random chance, and thus confirm the approach is feasible.

Model	Trainable parameters	Validation accuracy
Baseline (two fully-connected layers)	38.5M	77.1%
Primary (ResNet-18 + classifier head)	138K	91.6% (best 92.0%)

Table 1: model comparison

References

- [1] Grassknoted. Asl alphabet dataset. <https://www.kaggle.com/datasets/grassknoted/asl-alphabet>, 2017. Accessed: 2026-06-04.
- [2] Statistics Canada. Reports on disability and accessibility in canada. <https://www150.statcan.gc.ca/n1/en/catalogue/89-654-x>, 2023. Accessed: 2026-06-04.